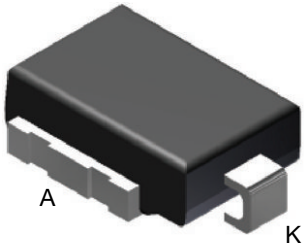
# Surface-Mount High Voltage Rectifiers




**DO-218 Compatible**

Anode ○—▶|—○ Cathode

| PRIMARY CHARACTERISTICS | |
|---|---|
| $I_{F(AV)}$ | 45 A |
| $V_{RRM}$ | 1200 V |
| $I_{FSM}$ | 700 A |
| $V_F$ at $I_F$ = 45 A ($T_J$ = 125 °C) | 1.05 V |
| $I_R$ | 10 µA |
| $E_{AS}$ | 20 mJ |
| $T_J$ max. | 175 °C |
| Package | DO-218AC |
| Circuit configurations | Single |

## FEATURES

AUTOMOTIVE GRADE Available

RoHS COMPLIANT HALOGEN FREE

- Excellent heat dissipation
- Oxide planar chip junction
- High surge current capability
- Ultra-low forward conduction
- High junction temperature capability
- High ESD capability
- High avalanche capability
- Meets MSL level 1, per J-STD-02, LF maximum peak of 245 °C
- AEC-Q101 qualified
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

## TYPICAL APPLICATIONS

- Fly-wheeling diode for big power motor in EV/HEV
- Single or three phase bridge rectification circuit
- High voltage block diode

## MECHANICAL DATA

**Case:** DO-218AC
Molding compound meets UL 94 V-0 flammability rating

Base P/N-M3 - halogen-free, RoHS-compliant, and commercial grade

Base P/NHM3_X - halogen-free, RoHS-compliant, and AEC-Q101 qualified

("X" denotes revision code e.g, A, B,...)

**Terminals:** matte tin plated leads, solderable per J-STD-002 and JESD 22-B102

M3 suffix meets JESD 201 class 2 whisker test, HM3 suffix meets JESD 201 class 2 whisker test

**Polarity:** as marked

**MAXIMUM RATINGS** ($T_A$ = 25 °C unless otherwise noted)

| PARAMETER | SYMBOL | SE45124 | UNIT |
|---|---|---|---|
| Device marking code | | SE45124 | |
| Maximum repetitive peak reverse voltage | $V_{RRM}$ | 1200 | V |
| Maximum DC forward current | $I_F$ (1) | 45 | A |
| | $I_F$ (2) | 4.2 | |
| Peak forward surge current 10 ms single half sine-wave superimposed on rated load | $I_{FSM}$ | 700 | A |
| 8 x 20 µs wave form by 10 surge pulses in 10 minutes | $I_{FSM}$ | 3500 | A |
| Typical Non-repetitive Avalanche energy at $I_{AS}$ = 1A, $T_J$ = 25 °C | $E_{AS}$ | 20 | mJ |
| Operating junction and storage temperature range | $T_J$, $T_{STG}$ | -55 to +175 | °C |

**Notes**
(1) Mounted on aluminum PCB 30 mm x 30 mm with aluminum heatsink
(2) Free air, mounted on recommended copper pad area


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| ELECTRICAL CHARACTERISTICS ($T_J$ = 25 °C unless otherwise noted) | | | | | | |
|---|---|---|---|---|---|---|
| PARAMETER | TEST CONDITIONS | | SYMBOL | TYP. | MAX. | UNIT |
| Instantaneous forward voltage | $I_F$ = 22.5 A | $T_J$ = 25 °C | $V_F$ (1) | 1.0 | - | V |
| | $I_F$ = 45 A | | | 1.13 | 1.3 | |
| | $I_F$ = 22.5 A | $T_J$ = 125 °C | | 0.90 | - | |
| | $I_F$ = 45 A | | | 1.05 | - | |
| Reverse current | Rated $V_R$ | $T_J$ = 25 °C | $I_R$ (2) | - | 10 | µA |
| | | $T_J$ = 125 °C | | 30 | - | |
| Typical junction capacitance | 400 V, 1 MHz | | $C_J$ | 35 | - | pF |

**Notes**
(1) Pulse test: 300 µs pulse width, 1 % duty cycle
(2) Pulse test: Pulse width ≤ 40 ms

| THERMAL CHARACTERISTICS ($T_A$ = 25 °c unless otherwise noted) | | | |
|---|---|---|---|
| PARAMETER | SYMBOL | SE45124 | UNIT |
| Typical thermal resistance | $R_{\theta JA}$ (1)(2) | 57 | °C/W |
| | $R_{\theta JM}$ (3) | 0.2 | |

**Notes**
(1) The heat generated must be less than the thermal conductivity from junction-to-ambient: $dP_D/dT_J < 1/R_{\theta JA}$
(2) Free air, mounted on recommended PCB, 2 oz. pad area; thermal resistance $R_{\theta JA}$ - junction to ambient
(3) Thermal resistance junction-to-mount to follow JEDEC® 51-14 transient dual interface test method (TDIM)

| IMMUNITY TO ELECTRICAL STATIC DISCHARGE TO THE FOLLOWING STANDARDS ($T_A$ = 25 °C unless otherwise noted) | | | | | |
|---|---|---|---|---|---|
| STANDARD | TEST TYPE | TEST CONDITIONS | SYMBOL | CLASS | VALUE |
| AEC-Q101-001 | Human body model (contact mode) | C = 100 pF, R = 1.5 kΩ | $V_C$ | H3B | > 8 kV |
| IEC 61000-4-2 (2) | Human body model (air discharge mode) (1) | C = 150 pF, R = 330 Ω | | 4 | > 30 kV |

**Notes**
(1) Immerse to IEC 61000-4-2 air discharge mode has a typical performance > 30 kV
(2) System ESD standard

| ORDERING INFORMATION (Example) | | | | |
|---|---|---|---|---|
| PREFERRED P/N | UNIT WEIGHT (g) | PREFERRED PACKAGE CODE | BASE QUANTITY | DELIVERY MODE |
| SE45124-M3/I | 2.56 | I | 750/reel | 13" diameter plastic tape and reel |
| SE45124HM3_A/I (1) | 2.56 | I | 750/reel | 13" diameter plastic tape and reel |

**Note**
(1) AEC-Q101 qualified


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## RATINGS AND CHARACTERISTICS CURVES ($T_A$ = 25 °C unless otherwise noted)

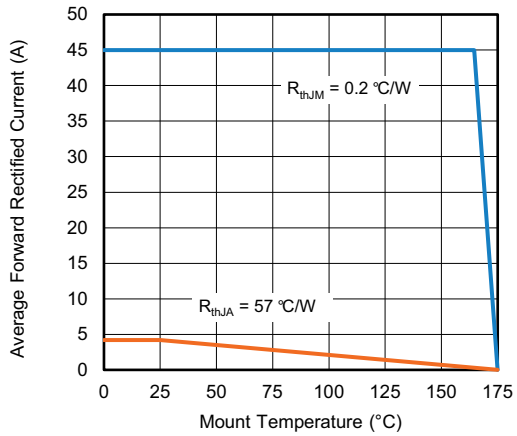



Fig. 1 - Forward Current Derating Curve

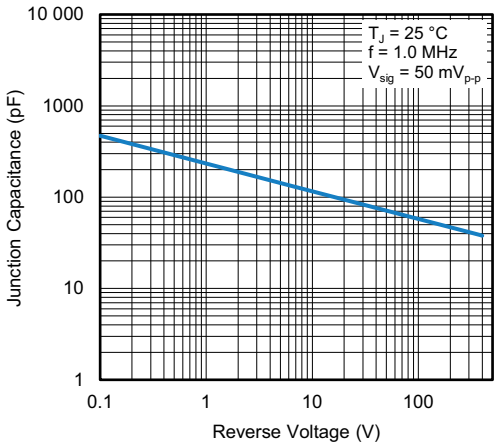



Fig. 4 - Typical Junction Capacitance

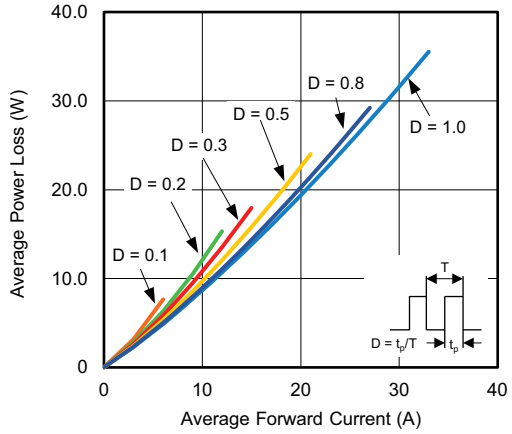



Fig. 2 - Forward Power Loss Characteristics

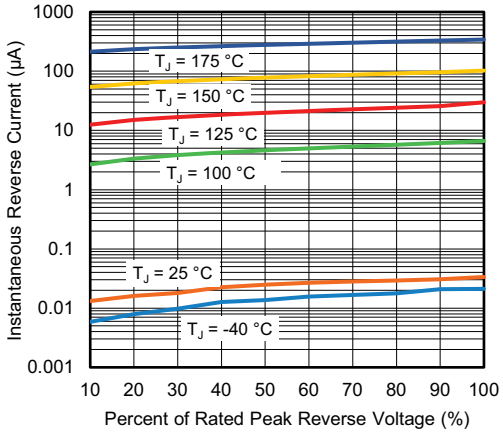



Fig. 5 - Typical Reverse Leakage Characteristics

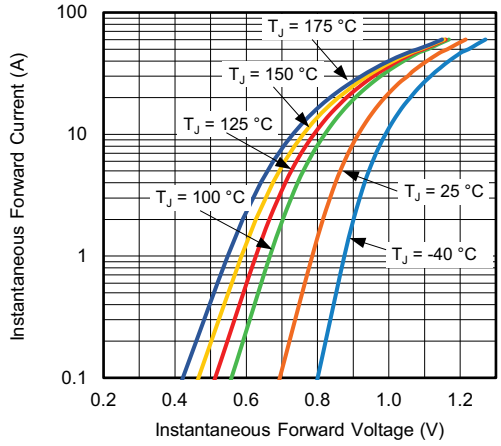



Fig. 3 - Typical Instantaneous Forward Characteristics

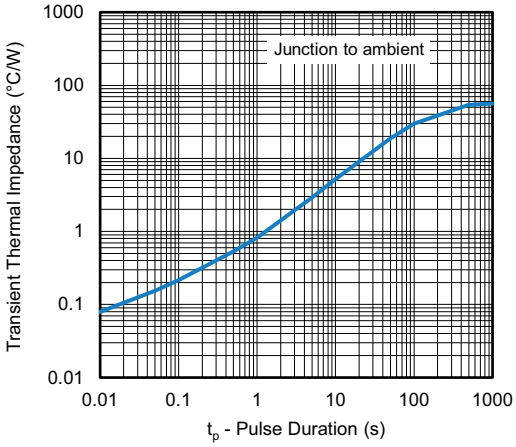
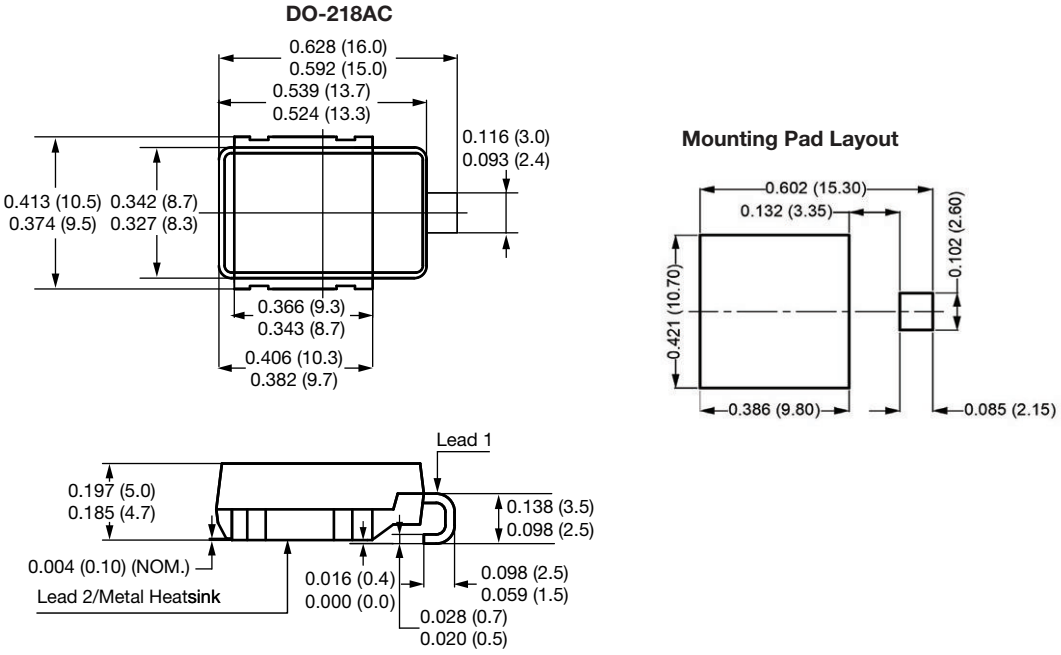



Fig. 6 - Typical Transient Thermal Impedance


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## PACKAGE OUTLINE DIMENSIONS in inches (millimeters)




**Note**

- Footprint in accordance with IPC 7351 standard


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